

One diagnostic. One page.

Purpose, inputs, evidence and workflow — compressed into editable cards and pills.

20 numbered diagnostics

1 supporting analysis

Detail + sources in notes

Repository baseline 0.5.2

IMPLEMENTED

D02 Target separation02

D04 Business rules · paused03

D06 Row completeness04

D08 Value semantics05

D11 Directional consistency06

D14 Population stability07

SUP Missingness Mechanism08

REGISTERED · PLANNED

D12 Label consistency09

D17 Resolution rate10

D20 AI proposals11

READ EVERY PAGE THE SAME WAY

Who / when → Inputs → Evidence
Processing → Issue gate → RCA

Planned flows are explicitly marked.
RCA closure does not certify a source fix.

DEFERRED · PROPOSED

D01 Look-ahead fields12

D03 Categorical leakage13

D05 Derivation identity14

D07 Disguised missingness15

D09 Robust-Z outliers16

D10 Boundary pile-up17

D13 Target-rate breaks18

D15 KS distance19

D16 Segment coverage20

D18 Resolved vs all21

D19 Seasoning profile22

DATA WORKBENCH / Diagnostic quick reference

Companion to the full 87-slide presentation

01 / 22

Single-feature target separation

Find unusually predictive features and weak individual signals before modeling.

USER & TRIGGER

Analyst / model team + SME
Run before feature selection, after target changes or when leakage is suspected.

INPUTS & READINESS

Ready snapshot; confirmed target, target type and event class; selected features; approved bin constraints.

OUTPUTS & EVIDENCE

ROC / AUC / Gini, fine and coarse bins, WOE / IV, feature categories and review candidates. Partial feature errors stay explicit.

PROCESS & LAUNCH

01 Select

Snapshot + features

02 Confirm

Target + event class

03 Freeze

Run / bin settings

04 Calculate

ROC + WOE / IV

05 Review

Final feature report

FINDING → ISSUE

Suspicious → Strong → Medium → Weak precedence. AUC cutoffs .90 / .70 / .55; IV .50 / .30 / .05 are tunable. Candidate thresholds are separate; user confirms or dismisses.

RCA → REMEDIATION

Check target construction, feature availability and bin evidence. Approve a root cause or mark unresolved; correct source / feature design externally, then upload and rerun.

A strong single feature is a review signal, not proof of leakage. Streaming previews are not final results.

Cross-field business rule

Check whether related values satisfy approved business rules, conditions and exceptions.

USER & TRIGGER

Data team / diagnostic owner + SME
Use for cross-field contradictions after rule selection and binding are approved.

INPUTS & READINESS

Ready data; published knowledge-base rules; use case; field-role bindings; rule-specific tolerances and exceptions.

OUTPUTS & EVIDENCE

PASS / VIOLATION / NOT-APPLICABLE; evaluated, exception and skip counts; violation rates, patterns and report evidence.

PROCESS & LAUNCH

CODE PATH SHOWN · LAUNCH PAUSED

01 Select rules

Approved KB release



02 Bind roles

Resolve required fields



03 Apply scope

Conditions / exceptions



04 Evaluate

Rate vs tolerance



05 Persist

Report + issue handoff

FINDING → ISSUE

If executed, VIOLATION auto-opens an issue in the runner. Rate $\leq$ tolerance is PASS. Missing roles, empty scope or entirely skipped rows are NOT-APPLICABLE.

RCA → REMEDIATION

Review mapping, exceptions and source facts versus rule design. Resolve RCA or mark unresolved; use an external data fix or governed rule change, then retest when enabled.

Current register pauses launch (workflow_pending). The processing and issue behavior shown exists in code.

Row-completeness reconciliation

Detect missing periods, duplicate pairs and gaps within the observed facility panel.

USER & TRIGGER

Data / analytics team
Run on a periodic panel to assess continuity before downstream analysis.

INPUTS & READINESS

One ready facility-period panel; facility ID, reporting period and confirmed grain; optional segment; continuity floor (default 95%).

OUTPUTS & EVIDENCE

Six rule outcomes: identifiers, period sequence, rows per period, duplicate pairs, facility gaps and segment gaps; masked failure evidence.

PROCESS & LAUNCH

01 Validate

Keys + panel grain



02 Define span

Observed first to last



03 Build grid

Required pairs



04 Reconcile

Gaps + duplicates



05 Review

Six rules + artifacts

FINDING → ISSUE

Continuity = received required pairs / required pairs. Review rule and segment failures even when the aggregate passes. User confirms a finding before issue creation.

RCA → REMEDIATION

Inspect invalid keys, calendar gaps, duplicate types and delivery gaps. Approve root cause or mark unresolved; correct the panel externally, upload new data and rerun.

Observed-span coverage only: cannot detect wholly absent facilities or missing rows outside observed endpoints.

Value-semantics classification

Distinguish censored, stale or inapplicable values before interpreting data-quality signals.

USER & TRIGGER

Analyst + domain SME
Acknowledge scope, confirm field roles and resolve prerequisites before execution.

INPUTS & READINESS

Ready table; selected fields and canonical roles; KB v0.2 / terminology v0.3; sentinel, date, window or panel declarations as applicable.

OUTPUTS & EVIDENCE

CENSORED / STALE_FROZEN / NOT_APPLICABLE tags. Four artifacts: bindings JSON, tags Parquet, ledger Parquet and report JSON.

PROCESS & LAUNCH

01 Acknowledge

About + table context



02 Bind roles

Confirm ranked matches



03 Check inputs

Required declarations



04 Freeze

Approved run context



05 Classify

Tags + evidence ledger

FINDING → ISSUE

Stale / frozen and populated not-applicable observations can form grouped candidates. A user confirms or dismisses them. NO_TAG, UNCLASSIFIED and UNSCOPED are coverage states.

RCA → REMEDIATION

Review source conventions and binding evidence. Approve root cause or mark unresolved; correct source data or declarations and rerun. This diagnostic never imputes or edits source values.

General context is valid; AI assists metadata only. Missing optional prerequisites are UNSCOPED; unresolved precedence fails closed.

Directional / monotonic consistency

Compare expected feature direction with empirical behavior against the target or approved reference.

USER & TRIGGER

Model / analytics team + SME
Use when business expectations and observed relationships may disagree.

INPUTS & READINESS

Numeric features; saved target or approved run reference; event / risk orientation; confirmed KB or manual expected directions.

OUTPUTS & EVIDENCE

Binned averages, Spearman and standardized univariate regression; direction / strength, charts and feature evidence. Pearson is display-only.

PROCESS & LAUNCH

01 Choose scope

Overall first run



02 Set reference

Target + risk meaning



03 Confirm

Expected directions



04 Freeze

Manifest + proposals



05 Compare

Three-signal evidence

FINDING → ISSUE

3/3 agreement is strong; 2/3 is moderate, subject to guards. Weak, conflicting or review-recommended findings need user assessment and confirmation before issue creation.

RCA → REMEDIATION

Check expectation, target orientation and population mix. Approve root cause or mark unresolved; correct source or govern a KB change. KB proposals require review and later curation.

First run is overall only; later runs may use one PSI-based segment. The reference complement is not analysed.

Population Stability Index (PSI)

Measure which feature distributions changed, and identify the bins driving the shift.

USER & TRIGGER

Analytics / model team
Compare populations after a data refresh, sample change or population shift.

INPUTS & READINESS

One snapshot split into Baseline / Current, or two compatible snapshots; selected features; optional target; approved bin definitions.

OUTPUTS & EVIDENCE

PSI and bin-level counts, shares and contributions; population / bin lineage and final report. Missing and unseen categories are explicit.

PROCESS & LAUNCH

01 Define

Baseline + Current



02 Route bins

Target-free or D02



03 Review

Approve + freeze bins



04 Calculate

PSI contributions



05 Investigate

Final shift evidence

FINDING → ISSUE

$PSI = \sum (Current - Baseline) \times \ln(Current / Baseline)$, with zero-share smoothing. Default bands: $<.10$ stable; $\geq .10$ watch; $\geq .25$ investigate. User reviews and confirms findings.

RCA → REMEDIATION

Inspect population choices, bin drivers and legitimate business changes. Approve root cause or mark unresolved; correct the source or assessment setup and manually reassess.

Bin reuse needs an exact promoted universal chain for two snapshots; split runs regenerate. Previews are not final.

Missingness Mechanism

Explain where missing values cluster and which observed patterns warrant investigation.

USER & TRIGGER

Analyst / data owner + SME
Investigate missingness after profiling or a completeness concern.

INPUTS & READINESS

Ready table; confirmed missing-value codes and analysis parameters; period context where relevant; frozen snapshot and seed.

OUTPUTS & EVIDENCE

Missingness masks, co-missing blocks, seven classification gates and shallow-tree explanations; immutable missingness_report artifact.

PROCESS & LAUNCH

01 Configure

Missing codes + scope



02 Freeze

Snapshot + seed



03 Map

Masks + co-missingness



04 Classify

Gates + shallow trees



05 Review

Immutable report

FINDING → ISSUE

Above-tolerance observations enter user review. Confirm with rationale to create an issue, or dismiss. Untestable results must remain distinct from “no pattern found.”

RCA → REMEDIATION

Inspect source capture, process and group / period patterns through shared RCA. Approve root cause or mark unresolved; remediate externally and manually reassess new data.

Descriptive evidence does not establish MCAR, MAR, MNAR or causality. Zero is valid unless explicitly declared missing.

Label-consistency rule

Identify target labels that contradict approved process or business definitions.

USER & TRIGGER

Target owner / SME + data team
Intended before model development or after a label-definition change.

INPUTS & READINESS

Proposed: labels, process-state fields, approved labeling rules, exceptions, scope and tolerances.

OUTPUTS & EVIDENCE

Proposed: label-rule conflicts, violation counts / rates and affected-record evidence. No current executor or artifact contract.

PROPOSED PROCESSING FLOW

01 Define

Label / process rules



02 Bind

Labels + process facts



03 Evaluate

Rule contradictions



04 Summarize

Conflicts + rates



05 Review

Owner / SME decision

PROPOSED FINDING → ISSUE

Proposed gate: compare deterministic rule violations with approved tolerance; have the label owner confirm whether each pattern is a defect before issue promotion.

PROPOSED RCA → ACTION

Proposed: separate source labeling errors from definition or timing ambiguity. Agree source / rule correction and a fresh assessment; diagnostic-specific automation is not implemented.

Registered as workflow_pending; planned method only. The numbered diagnostic has no current executable workflow.

Resolution / maturity rate by vintage

Show whether each vintage has enough resolved outcomes for the intended assessment.

USER & TRIGGER

Outcome / model owner + SME
Intended when recent cohorts may still contain many unresolved cases.

INPUTS & READINESS

Proposed: vintage, resolved / open status, observation cut-off, relevant horizon and calibrated maturity threshold.

OUTPUTS & EVIDENCE

Proposed: resolved / total rate by vintage, denominators and below-threshold cohorts. No current executor or artifact contract.

PROPOSED PROCESSING FLOW

01 Define

Vintages + cut-off



02 Classify

Resolved vs open



03 Measure

Resolved / total



04 Compare

Calibrated threshold



05 Review

Cohort suitability

PROPOSED FINDING → ISSUE

Proposed gate: flag low-resolution cohorts for contextual review. Calibrate thresholds to use case and horizon; an immature vintage is not automatically a data defect.

PROPOSED RCA → ACTION

Proposed: investigate observation horizon, closure timing and missing status updates. Agree monitoring, data correction or cohort selection changes; reassess with later data.

Registered as workflow_pending. Maturity is outcome- and horizon-specific; no current D17 execution workflow.

AI-proposal workflow

Turn data observations into reviewed knowledge-base proposals and reusable checks.

USER & TRIGGER

Analyst / SME + KB editor and reviewer
Intended when existing rules leave coverage gaps.

INPUTS & READINESS

Proposed: profile and context, existing KB content, proposal criteria and reviewer decisions. Confidence is a ranking aid.

OUTPUTS & EVIDENCE

Proposed: ranked suggestions, SME disposition and accepted KB content; approved checks route into the T2 rule engine.

PROPOSED PROCESSING FLOW

01 Profile

Data + KB context



02 Propose

AI-ranked suggestions



03 Validate

SME decisions



04 Publish

Governed KB content



05 Route

Accepted T2 checks

PROPOSED FINDING → ISSUE

Proposed gate: a suggestion does not directly become an issue. Humans validate meaning and publish accepted rules; executable diagnostics then produce reviewable findings.

PROPOSED RCA → ACTION

Proposed: retain proposal rationale and reviewer decisions. Route supported findings into issue / RCA handling. D11 proposal building blocks do not constitute the full D20 workflow.

Registered as workflow_pending. No fixed D20 executor or diagnostic metric; full workflow remains planned.

Post-outcome / look-ahead field

Find features that would not have been available at the prediction or assessment date.

USER & TRIGGER

Model developer / data owner
Intended before feature approval and
after source or transformation changes.

INPUTS & READINESS

Proposed: feature inventory, actual
availability timestamps, as-of date,
outcome date and transformation
lineage.

OUTPUTS & EVIDENCE

Proposed: flagged post-outcome /
look-ahead fields and timing evidence for
each field.

PROPOSED PROCESSING FLOW

01 Inventory

Features + lineage



02 Establish

Actual availability



03 Align

As-of / outcome dates



04 Flag

Look-ahead exposure



05 Review

Timing with owners

PROPOSED FINDING → ISSUE

Proposed gate: flag any post-outcome field included in features.
Validate actual operational availability and construction history before
treating it as leakage.

PROPOSED RCA → ACTION

Proposed: trace late source updates and look-ahead joins; rebuild
time-correct features or remove affected inputs, then reassess on a
valid snapshot.

Deferred catalog diagnostic: not registered or executable. Timestamp lineage alone may not prove availability.

Categorical leakage: association / purity

Find category values that act as near-deterministic proxies for the target.

USER & TRIGGER

Model / analytics team + SME
Intended before categorical feature acceptance.

INPUTS & READINESS

Proposed: categorical features, target and event class, base event rate, minimum category count and tuned cutoffs.

OUTPUTS & EVIDENCE

Proposed: category event rates versus base rate, volume, Cramér's V / IV and suspect feature-category pairs.

PROPOSED PROCESSING FLOW

01 Select

Categories + target



02 Gate volume

Minimum category count



03 Measure

Purity / association



04 Flag

Tuned effect sizes



05 Review

Timing + meaning

PROPOSED FINDING → ISSUE

Proposed gate: near 0 / 1 outcome rates (for example >~95%) far from base rate, or V / IV >~.5, subject to minimum volume. User / SME confirms leakage concern.

PROPOSED RCA → ACTION

Proposed: trace proxy codes, post-outcome assignment and legitimate segmentation. Correct or exclude leakage constructions and reassess on a time-valid sample.

Deferred; no executor. Use effect size and volume, not chi-square significance that over-fires at large sample sizes.

Derivation identity

Check whether a recorded derived field matches its approved calculation.

USER & TRIGGER

Data engineer / business owner
Intended after calculation, mapping or transformation changes.

INPUTS & READINESS

Proposed: recorded derived field,
component fields, approved formula,
units, rounding rules and tolerance.

OUTPUTS & EVIDENCE

Proposed: recomputed value, absolute
difference, tolerance breaches and
record-level reconciliation evidence.

PROPOSED PROCESSING FLOW

01 Approve

Formula + tolerance



02 Bind

Recorded + inputs



03 Recompute

Expected derivation



04 Compare

Absolute difference



05 Review

Mismatch evidence

PROPOSED FINDING → ISSUE

Proposed gate: $|\text{recorded} - \text{derived}| > \text{approved tolerance}$. Confirm formula version, units and rounding before promoting a mismatch to an issue.

PROPOSED RCA → ACTION

Proposed: locate incorrect source components, formula drift or pipeline errors. Correct the calculation / source, reload data and reconcile again.

Deferred catalog diagnostic: not registered or executable. Formula and tolerance must be explicitly approved.

Disguised missingness / staleness

Detect normally changing fields that become frozen or dominated by sentinel values.

USER & TRIGGER

Data owner / analyst + SME
Intended when refreshed data appears unexpectedly constant.

INPUTS & READINESS

Proposed: fields expected to vary, period and segment keys, sentinel declarations and a meaningful history window.

OUTPUTS & EVIDENCE

Proposed: within-segment variance over time, sentinel shares and affected segment-period patterns.

PROPOSED PROCESSING FLOW

01 Scope

Expected-varying fields



02 Declare

Sentinels + periods



03 Measure

Variance / sentinel share



04 Flag

Frozen patterns



05 Review

Business explanation

PROPOSED FINDING → ISSUE

Proposed gate: zero / near-zero variance where variation is expected, or excessive sentinel share. Confirm legitimate constants and missing-value conventions with the SME.

PROPOSED RCA → ACTION

Proposed: investigate default-filled fields, stale joins and feed refresh failures. Correct capture / delivery or metadata, then reassess updated periods.

Deferred; no D07 executor. Overlap with implemented D08 semantics does not make this catalog diagnostic active.

Statistical outlier: robust Z

Find improbable extremes in material numeric fields after checking value meaning.

USER & TRIGGER

Analyst / domain SME
Intended for tail review before modeling
or business use.

INPUTS & READINESS

Proposed: material numeric fields,
approved scope and value semantics;
robust-Z cutoff and contextual review
criteria.

OUTPUTS & EVIDENCE

Proposed: median, median absolute
deviation (MAD), robust-Z scores and
candidate extreme records.

PROPOSED PROCESSING FLOW

01 Select

Material fields



02 Gate

Value semantics



03 Estimate

Median + MAD



04 Score

Robust Z



05 Review

Extreme candidates

PROPOSED FINDING → ISSUE

Proposed score: $0.6745 \times (x - \text{median}) / \text{MAD}$. An example cutoff is $|Z| > \sim 3.5$. Confirm semantics and business plausibility; legitimate tails are not automatically defects.

PROPOSED RCA → ACTION

Proposed: inspect units, capture mistakes and valid rare events.
Correct proven source errors; otherwise document the tail decision
and reassess treatment.

Deferred; no executor. Value-semantics gating is required; constant-field / zero-MAD handling needs specification.

Boundary / pile-up detection

Find implausible concentrations at the declared minimum or maximum of a bounded field.

USER & TRIGGER

Data / analytics team + SME
Intended when values may be clipped, capped or defaulted.

INPUTS & READINESS

Proposed: bounded fields, declared limits, relevant segments / periods and an approved boundary-share threshold.

OUTPUTS & EVIDENCE

Proposed: share and count at each bound, distribution context and affected fields / segments.

PROPOSED PROCESSING FLOW

01 Declare

Bounds + semantics



02 Scope

Fields / populations



03 Measure

Mass at each bound



04 Compare

Tuned share limit



05 Review

Cap vs data defect

PROPOSED FINDING → ISSUE

Proposed gate: boundary share > an approved X%. Review contractual caps, censoring and operational limits before confirming an anomalous pile-up.

PROPOSED RCA → ACTION

Proposed: trace clipping, default assignment and source limits. Correct transformation / capture errors or document a valid bound; rerun on updated data.

Deferred catalog diagnostic: not registered or executable. No universal threshold is supplied for boundary mass.

Target-rate break / change-point

Find abrupt target-rate changes that may reflect a definition or process change.

USER & TRIGGER

Target owner / model team
Intended after label-policy changes or unexpected time trends.

INPUTS & READINESS

Proposed: dated target outcomes and denominators, definition / process change dates, analysis windows and significance criteria.

OUTPUTS & EVIDENCE

Proposed: target-rate series, candidate change points, break statistics and alignment with definition dates.

PROPOSED PROCESSING FLOW

01 Build

Target-rate series



02 Annotate

Definition / policy dates



03 Test

Structural breaks



04 Align

Breaks with events



05 Review

Definition vs reality

PROPOSED FINDING → ISSUE

Proposed gate: a break exceeding the agreed statistical criterion triggers contextual review. Compare denominator, population and definition changes before issue promotion.

PROPOSED RCA → ACTION

Proposed: distinguish real business shifts from labeling changes, timing and source errors. Correct or harmonize affected labels and reassess comparable periods.

Deferred; no executor. A statistical break alone does not establish a definitional defect or its cause.

KS distance

Complement PSI with a direct measure of distribution difference from a reference.

USER & TRIGGER

Analytics / model team
Intended when assessing sample or population representativeness.

INPUTS & READINESS

Proposed: compatible sample / reference populations, comparable numeric fields, scope and approved distance threshold.

OUTPUTS & EVIDENCE

Proposed: Kolmogorov–Smirnov distance, empirical distribution comparison and candidate shifted features.

PROPOSED PROCESSING FLOW

01 Define

Sample + reference



02 Align

Fields + populations



03 Compare

Empirical distributions



04 Measure

KS distance



05 Review

PSI + business context

PROPOSED FINDING → ISSUE

Proposed gate: KS distance above a tuned threshold. Read alongside sample size, PSI and practical materiality before confirming a distribution concern.

PROPOSED RCA → ACTION

Proposed: inspect sampling, population mix, extraction and genuine market changes. Correct source / sample problems or document a legitimate shift, then reassess.

Deferred; no executor. Statistical screening can over-flag at large sample sizes; interpret practical effect size.

Segment coverage

Find live-population segments that are absent or too thin in the assessment sample.

USER & TRIGGER

Analytics / population owner
Intended before sample acceptance or model development.

INPUTS & READINESS

Proposed: reference and sample populations, consistent segment definitions, counts and minimum acceptable sample share.

OUTPUTS & EVIDENCE

Proposed: segment presence and share comparisons, missing / thin segments and coverage gaps.

PROPOSED PROCESSING FLOW

01 Define

Reference + sample

02 Align

Segment definitions

03 Count

Presence + shares

04 Flag

Missing / thin groups

05 Review

Coverage materiality

PROPOSED FINDING → ISSUE

Proposed gate: a segment exists in reference but is below the approved sample minimum. Confirm intended exclusions and segment mapping before issue promotion.

PROPOSED RCA → ACTION

Proposed: trace selection filters, missing deliveries and mapping gaps. Correct extraction or sampling decisions and verify coverage on a refreshed sample.

Deferred catalog diagnostic: not registered or executable. Reference quality and scope drive the comparison.

Resolved-only vs all differential

Quantify how much the assessment changes when unresolved cases are excluded.

USER & TRIGGER

Outcome / model owner + SME
Intended when resolved-only selection may bias an estimate.

INPUTS & READINESS

Proposed: full cohort, resolved / open status, defined estimator, comparable observation window and materiality threshold.

OUTPUTS & EVIDENCE

Proposed: resolved-only estimate, full-cohort estimate, denominators and the difference between them.

PROPOSED PROCESSING FLOW

01 Define

Cohort + estimator

02 Split

Resolved vs full cohort

03 Estimate

Comparable measures

04 Compare

Difference vs limit

05 Review

Selection sensitivity

PROPOSED FINDING → ISSUE

Proposed gate: differential above an agreed Y indicates possible selection bias. Review sign, magnitude and estimator assumptions before confirming the concern.

PROPOSED RCA → ACTION

Proposed: investigate resolution timing, exclusions and outcome capture. Adjust the assessment approach or fix status / source data, then reassess both populations.

Deferred; no executor. A differential signals sensitivity; it does not by itself establish a data defect.

Seasoning / maturity profile

Identify recent vintages that have not developed far enough for the chosen outcome horizon.

USER & TRIGGER

Outcome / model owner + SME
Intended before selecting cohorts for outcome estimation.

INPUTS & READINESS

Proposed: vintage / origination date, months on book or age, cumulative outcomes, observation cut-off and mature reference cohorts.

OUTPUTS & EVIDENCE

Proposed: cumulative outcome / seasoning curves by vintage and cohort maturity gaps at comparable ages.

PROPOSED PROCESSING FLOW

01 Define

Vintages + horizon



02 Align

Age / months on book



03 Profile

Cumulative outcomes



04 Compare

Mature seasoning



05 Review

Cohort readiness

PROPOSED FINDING → ISSUE

Proposed gate: recent vintages below the use-case maturity profile trigger review. Separate shorter observation windows from incomplete or late source capture.

PROPOSED RCA → ACTION

Proposed: inspect elapsed horizon, reporting delay and genuine cohort effects. Wait for maturation, revise eligible cohorts or correct source capture, then reassess.

Deferred catalog diagnostic: not registered or executable. Maturity thresholds depend on outcome and use case.